

Machine Learning in Python

Supervised Learning - Classification and Metrics

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Introduction to Classification

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- In **regression**, the target variable is continuous (e.g., predicting a price).

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- **Multiclass Classification**: The target variable has more than two classes (e.g., "cat", "dog", "weasel"). In this case, the model predicts one of several possible categories.

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- More advanced algorithms like **Random Forests**, **Gradient Boosting**, and **Neural Networks**.

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The sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where z is a linear combination of the input features.

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	Predicted = 1	Predicted = 0
Actual = 1	True Positive (TP)	False Negative (FN)
Actual = 0	False Positive (FP)	True Negative (TN)

From these four numbers we obtain the core metrics summarized next.

Accuracy (Overall Success Rate)

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Definition

Accuracy is a measure of how often the classifier is correct across all predictions. It answers the question: "What fraction of **all** predictions are correct?"

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

Despite its simplicity, accuracy can be misleading, especially in imbalanced datasets where one class dominates.

Precision (Positive Predictive Value)

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Definition

Precision is a measure of the accuracy of positive predictions. It answers the question: "When the classifier predicts 1, how often is it correct?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision indicates that when the model predicts a positive class, it is likely correct, but it does not account for false negatives.

Recall (Sensitivity, True-Positive Rate)

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Definition

Recall is a measure of the model's ability to capture all positive instances. It answers the question: "Of all the actual 1 cases, how many did we catch?"

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

High recall indicates that the model is good at identifying positive cases, but it does not consider false positives.

F1-Score

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Definition

F1-Score is the harmonic mean of precision and recall, providing a balance between the two metrics:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is particularly useful when the class distribution is imbalanced, as it considers both false positives and false negatives. High F1-Score indicates a good balance between precision and recall, while a low F1-Score suggests that either precision or recall (or both) are low.

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Precision-Recall Curve is a graphical representation of the trade-off between precision and recall for different threshold values.

Receiver Operating Characteristic (ROC) Curve

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ROC Curve is a graphical representation of a classifier's performance across different thresholds. It plots the true positive rate (recall) against the false positive rate (FPR) at various threshold settings.

Area Under the Curve (AUC)

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Area Under the Curve (AUC) is a single scalar value that summarizes the performance of a classifier across all possible thresholds. It is calculated as the area under the ROC curve. AUC provides an aggregate measure of performance across all classification thresholds, making it useful for comparing different classifiers.

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The decision boundary is implicitly defined by the k nearest points, allowing the model to capture highly non-linear class boundaries without explicit training.

Mathematical Formulation

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Definition

Given a training set $\{(x_i, y_i)\}_{i=1}^n$ where $y_i \in \{1, \dots, C\}$, the prediction for a query point $\hat{x}$ is:

$$\hat{y} = \text{mode}(\{y_j | x_j \in N_k(\hat{x})\}),$$

where $N_k(\hat{x})$ denotes the set of k training points closest to $\hat{x}$.

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 - Requires a good choice of distance metric.

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$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1, \quad \forall i,$$

where $\mathbf{w}$ is the weight vector, b is the bias term, and y_i is the class label of the i -th training example.

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Visualization should help. . .

Linear SVMs

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Linear SVMs are used when the data is linearly separable. The decision boundary is a hyperplane defined by the equation:

$$\mathbf{w}^T \mathbf{x} + b = 0,$$

where $\mathbf{w}$ is the weight vector and b is the bias term. The SVM finds the hyperplane that maximizes the margin between the two classes.

Non-Linear SVMs

Non-Linear SVMs

Definition

Non-Linear SVMs use kernel functions to transform the input space into a higher-dimensional space where a linear hyperplane can separate the classes. Common kernels include:

- **Polynomial Kernel**: Maps the input features into a polynomial feature space.
- **Radial Basis Function (RBF) Kernel**: Maps the input features into an infinite-dimensional space, allowing for complex decision boundaries.
- **Sigmoid Kernel**: Similar to the activation function in neural networks, it maps the input features into a space that can capture non-linear relationships.

Kernel Trick

Kernel Trick

Definition

The **Kernel Trick** allows SVMs to operate in high-dimensional feature spaces without explicitly computing the coordinates of the data in that space. Instead, it computes the inner products between the images of all pairs of data points in the feature space, which is computationally efficient.

This is done using a kernel function $K(\mathbf{x}_i, \mathbf{x}_j)$ that computes the inner product in the transformed space:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j),$$

where ϕ is the mapping function that transforms the input features into the higher-dimensional space.